

Techniques of Integration

Basic Integration Formulas

1. $\int du = u + C$
2. $\int k du = ku + C,$ (any number k)
3. $\int (du + dv) = \int du + \int dv$
4. $\int u^n du = \frac{u^{n+1}}{n+1} + C$ ($n \neq -1$)
5. $\int \frac{du}{u} = \ln|u| + C$
6. $\int \sin u du = -\cos u + C$
7. $\int \cos u du = \sin u + C$
8. $\int \sec^2 u du = \tan u + C$
9. $\int \csc^2 u du = -\cot u + C$
10. $\int \sec u \tan u du = \sec u + C$
11. $\int \csc u \cot u du = -\csc u + C$
12. $\int \tan u du = -\ln|\cos u| + C = \ln|\sec u| + C$
13. $\int \cot u du = \ln|\sin u| + C = -\ln|\csc u| + C$
14. $\int e^u du = e^u + C$
15. $\int a^u du = \frac{a^u}{\ln a} + C$ ($a > 0, a \neq 1$)
16. $\int \sinh u du = \cosh u + C$
17. $\int \cosh u du = \sinh u + C$
18. $\int \frac{du}{\sqrt{a^2 - u^2}} = \sin^{-1}\left(\frac{u}{a}\right) + C$
19. $\int \frac{du}{a^2 + u^2} = \frac{1}{a} \tan^{-1}\left(\frac{u}{a}\right) + C$

$$20. \int \frac{du}{u\sqrt{u^2-a^2}} = \frac{1}{a} \sec^{-1} \left| \frac{u}{a} \right| + C$$

$$21. \int \frac{du}{\sqrt{a^2+u^2}} = \sinh^{-1} \left(\frac{u}{a} \right) + C \quad (a > 0)$$

$$22. \int \frac{du}{\sqrt{u^2-a^2}} = \cosh^{-1} \left(\frac{u}{a} \right) + C \quad (u > a > 0)$$

Making a simplifying substitution

Example 1: Evaluate the following integral

$$\int \frac{2x-9}{\sqrt{x^2-9x+1}} dx$$

Solution: Let: $u = x^2 - 9x + 1$, $du = 2x - 9$

$$\begin{aligned} \int \frac{2x-9}{\sqrt{x^2-9x+1}} dx &= \int \frac{du}{\sqrt{u}} = \int u^{-1/2} du \\ &= \frac{u^{-\frac{1}{2}+1}}{\left(-\frac{1}{2}\right)+1} + C \\ &= 2u^{1/2} + C = 2\sqrt{x^2-9x+1} + C \end{aligned}$$

Completing the Square

Example 2: Evaluate the following integral

$$\int \frac{dx}{\sqrt{8x-x^2}}$$

Solution: $8x - x^2 = -(x^2 - 8x) = -(x^2 - 8x + 16) + 16 = 16 - (x - 4)^2$

$$\begin{aligned} \int \frac{dx}{\sqrt{8x-x^2}} &= \int \frac{dx}{\sqrt{16-(x-4)^2}} = \int \frac{du}{\sqrt{a^2-u^2}} \\ &= \sin^{-1} \left(\frac{u}{a} \right) + C \\ &= \sin^{-1} \left(\frac{x-4}{4} \right) + C \end{aligned}$$

Expanding a Power and Using Trigonometric Identity

Example 3: Evaluate the following integral

$$\int (\sec x + \tan x)^2 dx$$

Solution:

$$\int (\sec x + \tan x)^2 dx = \int (\sec^2 x + 2 \sec x \tan x + \tan^2 x) dx$$

But, $\sec^2 x = \tan^2 x + 1$, therefore,

$$\begin{aligned} \int (\sec^2 x + 2 \sec x \tan x + \tan^2 x) dx &= \int (\sec^2 x + 2 \sec x \tan x + \sec^2 x - 1) dx \\ &= 2 \int \sec^2 x dx + 2 \int \sec x \tan x dx - \int 1 dx \\ &= 2 \tan x + 2 \sec x - x + C \end{aligned}$$

Eliminating a Square Root

Example 4: Evaluate the following integral

$$\int_0^{\pi/4} \sqrt{1 + \cos 4x} dx$$

Solution:

$$\cos^2 x = \frac{1 + \cos 2x}{2} \quad \text{or} \quad 1 + \cos 2x = 2 \cos^2 x$$

With the same identity the angle x can be doubled: $1 + \cos 4x = 2 \cos^2 2x$

$$\begin{aligned} \int_0^{\pi/4} \sqrt{1 + \cos 4x} dx &= \int_0^{\pi/4} \sqrt{2 \cos^2 2x} dx = \sqrt{2} \int_0^{\pi/4} |\cos 2x| dx \\ &= \sqrt{2} \left[\frac{\sin 2x}{2} \right]_0^{\pi/4} = \sqrt{2} \left[\frac{1}{2} - 0 \right] = \frac{1}{\sqrt{2}} \end{aligned}$$

Separating a fraction

Example 5: Evaluate the following integral

$$\int \frac{3x + 2}{\sqrt{1 - x^2}} dx$$

Solution:

$$\int \frac{3x + 2}{\sqrt{1 - x^2}} dx = 3 \int \frac{x dx}{\sqrt{1 - x^2}} + 2 \int \frac{dx}{\sqrt{1 - x^2}}$$

$$u = 1 - x^2, du = -2x dx \Rightarrow x dx = -\frac{1}{2} du$$

The first of these integrals will be solved as:

$$\begin{aligned} \therefore 3 \int \frac{x dx}{\sqrt{1 - x^2}} &= 3 \int \frac{\left(-\frac{1}{2}\right) du}{\sqrt{u}} = -\frac{3}{2} \int u^{-\frac{1}{2}} du \\ &= -\frac{3}{2} \left(\frac{u^{\frac{1}{2}}}{\frac{1}{2}} \right) + C_1 = -3\sqrt{1 - x^2} + C_1 \end{aligned}$$

The second of these integrals will be solved as:

$$2 \int \frac{dx}{\sqrt{1 - x^2}} = 2 \sin^{-1} x + C_2$$

The combination of these results and renaming $C_1 + C_2$ as C will give:

$$\int \frac{3x + 2}{\sqrt{1 - x^2}} dx = -3\sqrt{1 - x^2} + 2 \sin^{-1} x + C$$